MD HASANUR RAHMAN

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CURRENT POSITION

University of Iowa *2021-present*
Department of Computer Science
PhD Student

Argonne National Laboratory *2021-present*
Mathematics and Computer Science Division
Visiting Student

RESEARCH INTERESTS

HPC Fault Tolerance, Lossy Compression, Machine Learning Application

EDUCATION

University of Iowa *2021-present*
PhD in Computer Science
Advisor: Dr. Guanpeng Li

Bangladesh University of Engineering and Technology *2015-2019*
BSc in Computer Science and Engineering

PUBLICATIONS

Peppa-X: Finding Program Test Inputs to Bound Silent Data Corruption Vulnerability in HPC Applications
Md Hasanur Rahman, Aabid Shamji, Shengjian Guo, Guanpeng Li
ACM International Conference for High-Performance Computing, Networking, Storage and Analysis (SC'21) Acceptance rate: 23.6%

RESEARCH EXPERIENCE

Characterizing Error Propagation across Multiple Inputs in HPC Applications *2021*

- Developed an automated program analysis framework that depicts the upper bound of program SDC (silent data corruption) probability based on error propagation characteristics over multiple inputs.
- Proposed a LLVM-based dynamic input fuzzing technique that identify the SDC-bound inputs for benchmarking HPC resilience.

Optimizing Time Complexity to Search for Minimum Spanning Tree *2019*

- Proposed an efficient search technique that identifies the minimum spanning tree from a given weighted graph based on iteration-based reduction properties of graphs
- Reduced the time complexity of the search by a log factor in weighted graph benchmarks

WORK EXPERIENCE

University of Iowa *2021-present*
Research Assistant *Iowa*

Samsung Research *2019-2021*
Software Engineer *Bangladesh*

PROFESSIONAL SERVICE

Subreviewer	ISSRE (2022), MiddleWare (2022), HPDC (2022), SELSE (2022), PRDC (2021)
Mentored Students	Zhengyang He (2022)